

Base case: $n = 1$. Given any $s \geq 0$, $r(s, u) = s$ for all $s < u \leq c_0(s)$, implying

$$\text{MMD}^2(p_{s|s}^{\theta_0}(\mathbf{x}_s), p_{s|u}^{\theta_1^*}(\mathbf{x}_s)) \stackrel{(a)}{=} \text{MMD}^2(q_s(\mathbf{x}_s), p_{s|u}^{\theta_1^*}(\mathbf{x}_s)) \stackrel{(b)}{=} 0 \quad (47)$$

for $u \leq c_0(s)$ where (a) is implied by Lemma 4 and (b) is implied by Lemma 3.

Inductive assumption: $n - 1$. We assume $p_{s|u}^{\theta_{n-1}^*}(\mathbf{x}_s) = q_s(\mathbf{x}_s)$ for all $s \leq u \leq r_{n-2}^{-1}(s, c_0(s))$.

We inspect the target distribution $p_{s|r(s,u)}^{\theta_{n-1}^*}(\mathbf{x}_s)$ in Eq. (8) if optimized on $s \leq u \leq r_{n-1}^{-1}(s, c_0(s))$. On this interval, we can apply $r(s, \cdot)$ to the inequality and get $s = r(s, s) \leq r(s, u) \leq r(s, r_{n-1}^{-1}(s, c_0(s))) = r_{n-2}^{-1}(s, c_0(s))$ since $r(s, \cdot)$ is increasing. And by inductive assumption $p_{s|r(s,u)}^{\theta_{n-1}^*}(\mathbf{x}_s) = q_s(\mathbf{x}_s)$ for $s \leq r(s, u) \leq r_{n-2}^{-1}(s, c_0(s))$, this implies minimizing

$$\mathbb{E}_{s,u} \left[w(s, u) \text{MMD}^2(p_{s|r(s,u)}^{\theta_{n-1}^*}(\mathbf{x}_s), p_{s|u}^{\theta_n}(\mathbf{x}_s)) \right]$$

on $s \leq u \leq r_{n-1}^{-1}(s, c_0(s))$ is equivalent to minimizing

$$\mathbb{E}_{s,u} \left[w(s, u) \text{MMD}^2(q_s(\mathbf{x}_s), p_{s|u}^{\theta_n}(\mathbf{x}_s)) \right] \quad (48)$$

for $s \leq u \leq r_{n-1}^{-1}(s, c_0(s))$. Lemma 3 implies that its minimum achieves $p_{s|u}^{\theta_n}(\mathbf{x}_s) = q_s(\mathbf{x}_s)$.

Lastly, taking $n \rightarrow \infty$ implies $\lim_{n \rightarrow \infty} r_n^{-1}(s, c_0(s)) = 1$ and thus the induction covers the entire $[s, 1]$ interval given each s . Therefore, $\lim_{n \rightarrow \infty} \text{MMD}(q_s(\mathbf{x}_s), p_{s|t}^{\theta_n}(\mathbf{x}_s)) = 0$ for all $0 \leq s \leq t \leq 1$. $\square$

B.5. Self-Consistency Implies Marginal Preservation

Without assuming marginal preservation, it is important to define the marginal distribution of $\mathbf{x}_s$ under generalized interpolants $q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t)$ as

$$q_{s|t}(\mathbf{x}_s) = \iint q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t) q_t(\mathbf{x} | \mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x}_t d\mathbf{x} \quad (49)$$

and we show that with self-consistent interpolants, this distribution is invariant of t , i.e. $q_{s|t}(\mathbf{x}_s) = q_s(\mathbf{x}_s)$.

Lemma 5. *If the interpolant $q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t)$ is self-consistent, the marginal distribution $q_{s|t}(\mathbf{x}_s)$ as defined in Eq. (49) satisfies $q_s(\mathbf{x}_s) = q_{s|t}(\mathbf{x}_s)$ for all $t \in [s, 1]$.*

Proof. For $t \in [s, 1]$,

$$q_{s|t}(\mathbf{x}_s) = \iint q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t) q_t(\mathbf{x} | \mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x}_t d\mathbf{x} \quad (50)$$

$$= \iint q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t) q_t(\mathbf{x}_t) \int \frac{q_t(\mathbf{x}_t | \mathbf{x}, \epsilon) q(\mathbf{x}) p(\epsilon)}{q_t(\mathbf{x}_t)} d\epsilon d\mathbf{x}_t d\mathbf{x} \quad (51)$$

$$= \iint q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t) q(\mathbf{x}) \int q_t(\mathbf{x}_t | \mathbf{x}, \epsilon) p(\epsilon) d\epsilon d\mathbf{x}_t d\mathbf{x} \quad (52)$$

$$= \iint q(\mathbf{x}) p(\epsilon) \left(\int q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t) q_t(\mathbf{x}_t | \mathbf{x}, \epsilon) d\mathbf{x}_t \right) d\epsilon d\mathbf{x} \quad (53)$$

$$= \iint q(\mathbf{x}) p(\epsilon) \left(\int q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t) q_{t,1}(\mathbf{x}_t | \mathbf{x}, \epsilon) d\mathbf{x}_t \right) d\epsilon d\mathbf{x} \quad (54)$$

$$\stackrel{(a)}{=} \iint q_{s,1}(\mathbf{x}_s | \mathbf{x}, \epsilon) q(\mathbf{x}) p(\epsilon) d\epsilon d\mathbf{x} \quad (55)$$

$$\stackrel{(b)}{=} \iint q_s(\mathbf{x}_s | \mathbf{x}, \epsilon) q(\mathbf{x}) p(\epsilon) d\epsilon d\mathbf{x} \quad (56)$$

$$= q_s(\mathbf{x}_s) \quad (57)$$

where (a) uses definition of self-consistent interpolants and (b) uses definition of our generalized interpolant. $\square$

We show in Appendix C.1 that DDIM is an example self-consistent interpolant. Furthermore, DDPM posterior (Ho et al., 2020; Kingma et al., 2021) is also self-consistent (see Lemma 6).

B.6. Existence of Deterministic Minimizer

We present the formal statement for the deterministic minimizer.

Proposition 3. *If for all $t \in [0, 1]$, $s \in [0, t]$, $\gamma_{s|t} \equiv 0$, $\mathbf{I}_{s|t}(\mathbf{x}, \mathbf{x}_t)$ is invertible w.r.t. $\mathbf{x}$, and there exists $C_1 < \infty$ such that $\|\mathbf{I}_{t|1}(\mathbf{x}, \epsilon)\| < C_1\|\mathbf{x} - \epsilon\|$, then there exists a function $h_{s|t} : \mathbb{R}^D \rightarrow \mathbb{R}^D$ such that*

$$q_s(\mathbf{x}_s) = \iint q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) \delta(\mathbf{x} - h_{s|t}(\mathbf{x}_t)) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t. \quad (58)$$

Proof. Let $\mathbf{I}_{s|t}^{-1}(\cdot, \mathbf{x}_t)$ be the inverse of $\mathbf{I}_{s|t}$ w.r.t. $\mathbf{x}$ such that $\mathbf{I}_{s|t}^{-1}(\mathbf{I}_{s|t}(\mathbf{x}, \mathbf{x}_t), \mathbf{x}_t) = \mathbf{x}$ and $\mathbf{I}_{s|t}(\mathbf{I}_{s|t}^{-1}(\mathbf{y}, \mathbf{x}_t), \mathbf{x}_t) = \mathbf{y}$ for all $\mathbf{x}, \mathbf{x}_t, \mathbf{y} \in \mathbb{R}^D$. Since there exists $C_1 < \infty$ such that $\|\mathbf{I}_{t|1}(\mathbf{x}, \epsilon)\| < C_1\|\mathbf{x} - \epsilon\|$ for all $t \in [0, 1]$, the PF-ODE of the original interpolant $\mathbf{I}_{t|1}(\mathbf{x}, \epsilon) = \mathbf{I}_t(\mathbf{x}, \epsilon)$ exists for all $t \in [0, 1]$ (Albergo et al., 2023). Then, for all $t \in [0, 1]$, $s \in [0, t]$, we let

$$\hat{h}_{s|t}(\mathbf{x}_t) = \mathbf{x}_t + \int_t^s \mathbb{E}_{\mathbf{x}, \epsilon} \left[\frac{\partial}{\partial u} \mathbf{I}_u(\mathbf{x}, \epsilon) \Big| \mathbf{x}_u \right] du, \quad (59)$$

which pushes forward the measure $q_t(\mathbf{x}_t)$ to $q_s(\mathbf{x}_s)$. We define:

$$h_{s|t}(\mathbf{x}_t) = \mathbf{I}_{s|t}^{-1}(\hat{h}_{s|t}(\mathbf{x}_t), \mathbf{x}_t). \quad (60)$$

Then, since $\gamma_{s|t} \equiv 0$, $q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) = \delta(\mathbf{x}_s - \mathbf{I}_{s|t}(\mathbf{x}, \mathbf{x}_t))$ where $\mathbf{x} \sim \delta(\mathbf{x} - h_{s|t}(\mathbf{x}_t))$. Therefore,

$$\mathbf{x}_s = \mathbf{I}_{s|t}(h_{s|t}(\mathbf{x}_t), \mathbf{x}_t) = \mathbf{I}_{s|t}(\mathbf{I}_{s|t}^{-1}(\hat{h}_{s|t}(\mathbf{x}_t), \mathbf{x}_t), \mathbf{x}_t) = \hat{h}_{s|t}(\mathbf{x}_t) \quad (61)$$

whose marginal follows $q_s(\mathbf{x}_s)$ due to it being the result of PF-ODE trajectories starting from $q_t(\mathbf{x}_t)$. $\square$

Concretely, DDIM interpolant satisfies all of the deterministic assumption, the regularity condition, and the invertibility assumption because it is a linear function of $\mathbf{x}$ and $\mathbf{x}_t$. Therefore, any diffusion or FM schedule with DDIM interpolant will enjoy a deterministic minimizer $p_{s|t}^\theta(\mathbf{x}|\mathbf{x}_t)$.

C. Analysis of Simplified Parameterization

C.1. DDIM Interpolant

We check that DDIM interpolant is self-consistent. By definition, $q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) = \delta(\mathbf{x}_s - \text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, s))$. We check that for all $s \leq r \leq t$,

$$\begin{aligned} \int q_{s|r}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_r) q_{r|t}(\mathbf{x}_r|\mathbf{x}, \mathbf{x}_t) d\mathbf{x}_r &= \int \delta(\mathbf{x}_s - \text{DDIM}(\mathbf{x}_r, \mathbf{x}, r, s)) \delta(\mathbf{x}_r - \text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, r)) d\mathbf{x}_r \\ &= \delta(\mathbf{x}_s - \text{DDIM}(\text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, r), \mathbf{x}, r, s)) \end{aligned}$$

where

$$\begin{aligned} \text{DDIM}(\text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, r), \mathbf{x}, r, s) &= \alpha_s \mathbf{x} + (\sigma_s / \sigma_r) ([\alpha_r \mathbf{x} + (\sigma_r / \sigma_t)(\mathbf{x}_t - \alpha_t \mathbf{x})] - \alpha_r \mathbf{x}) \\ &= \alpha_s \mathbf{x} + (\sigma_s / \sigma_r) (\sigma_r / \sigma_t) (\mathbf{x}_t - \alpha_t \mathbf{x}) \\ &= \alpha_s \mathbf{x} + (\sigma_s / \sigma_t) (\mathbf{x}_t - \alpha_t \mathbf{x}) \\ &= \text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, s) \end{aligned}$$

Therefore, $\delta(\mathbf{x}_s - \text{DDIM}(\text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, r), \mathbf{x}, r, s)) = \delta(\mathbf{x}_s - \text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, s))$. So DDIM is self-consistent.

It also implies a Gaussian forward process $q_t(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\alpha_t \mathbf{x}, \sigma_t^2 \sigma_d^2 I)$ as in diffusion models. By definition,

$$q_{t,1}(\mathbf{x}_t|\mathbf{x}) = q_t(\mathbf{x}_t|\mathbf{x}) = \int q_t(\mathbf{x}_t|\mathbf{x}, \epsilon) p(\epsilon) d\epsilon$$

so that $\mathbf{x}_t$ is a deterministic transform given $\mathbf{x}$ and ϵ , i.e., $\mathbf{x}_t = \text{DDIM}(\epsilon, \mathbf{x}, t, 1) = \alpha_t \mathbf{x} + \sigma_t \epsilon$, which implies $q_t(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\alpha_t \mathbf{x}, \sigma_t^2 \sigma_d^2 I)$.

C.2. Reusing $\mathbf{x}_t$ for $\mathbf{x}_r$

We propose that instead of sampling $\mathbf{x}_r$ via forward flow $\alpha_r \mathbf{x} + \sigma_r \epsilon$ we reuse $\mathbf{x}_t$ such that $\mathbf{x}_r = \text{DDIM}(\mathbf{x}_t, \mathbf{x}, r, t)$ to reduce variance. In fact, for any self-consistent interpolant, one can reuse $\mathbf{x}_t$ via $\mathbf{x}_r \sim q_{r|t}(\mathbf{x}_r | \mathbf{x}, \mathbf{x}_t)$ and $\mathbf{x}_r$ will follow $q_r(\mathbf{x}_r)$ marginally. We check

$$q_r(\mathbf{x}_r) \stackrel{(a)}{=} q_{r|t}(\mathbf{x}_r) = \iint q_{r|t}(\mathbf{x}_r | \mathbf{x}, \mathbf{x}_t) q_t(\mathbf{x} | \mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t = \iint q_{r|t}(\mathbf{x}_r | \mathbf{x}, \mathbf{x}_t) \int q_t(\mathbf{x}_t | \mathbf{x}, \epsilon) q(\mathbf{x}) p(\epsilon) d\epsilon d\mathbf{x} d\mathbf{x}_t$$

where (a) is due to Lemma 5. We can see that sampling $\mathbf{x}, \mathbf{x}_t$ first then $\mathbf{x}_r \sim q_{r|t}(\mathbf{x}_r | \mathbf{x}, \mathbf{x}_t)$ respects the marginal distribution $q_r(\mathbf{x}_r)$.

C.3. Simplified Objective

We derive our simplified objective. Given MMD defined in Eq. (1), we write our objective as

$$\mathcal{L}_{\text{IMM}}(\theta) = \mathbb{E}_{s,t} \left[w(s, t) \left\| \mathbb{E}_{\mathbf{x}_t} [k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \cdot)] - \mathbb{E}_{\mathbf{x}_r} [k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r), \cdot)] \right\|_{\mathcal{H}}^2 \right] \quad (62)$$

$$\stackrel{(a)}{=} \mathbb{E}_{s,t} \left[w(s, t) \left\| \mathbb{E}_{\mathbf{x}_t, \mathbf{x}_r} [k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \cdot) - k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r), \cdot)] \right\|_{\mathcal{H}}^2 \right] \quad (63)$$

$$= \mathbb{E}_{s,t} \left[w(s, t) \left\langle \mathbb{E}_{\mathbf{x}_t, \mathbf{x}_r} [k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \cdot) - k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r), \cdot)], \mathbb{E}_{\mathbf{x}'_t, \mathbf{x}'_r} [k(\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t), \cdot) - k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r), \cdot)] \right\rangle \right] \quad (64)$$

$$= \mathbb{E}_{s,t} \left[w(s, t) \mathbb{E}_{\mathbf{x}_t, \mathbf{x}_r, \mathbf{x}'_t, \mathbf{x}'_r} \left[\langle k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \cdot), k(\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t), \cdot) \rangle + \langle k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r), \cdot), k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r), \cdot) \rangle \right. \right. \\ \left. \left. - \langle k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \cdot), k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r), \cdot) \rangle - \langle k(\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t), \cdot), k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r), \cdot) \rangle \right] \right] \quad (65)$$

$$= \mathbb{E}_{\mathbf{x}_t, \mathbf{x}_r, \mathbf{x}'_t, \mathbf{x}'_r, s, t} \left[w(s, t) \left[k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)) + k(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r), \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r)) \right. \right. \\ \left. \left. - k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r)) - k(\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t), \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r)) \right] \right] \quad (66)$$

where $\langle \cdot, \cdot \rangle$ is in RKHS, (a) is due to the correlation between $\mathbf{x}_r$ and $\mathbf{x}_t$ by re-using $\mathbf{x}_t$.

C.4. Empirical Estimation

As proposed in Gretton et al. (2012), MMD is typically estimated with V-statistics by instantiating a matrix of size $M \times M$ such that a batch of B $\mathbf{x}$ samples, $\{x^{(i)}\}_{i=1}^B$, is separated into groups of M (assume B is divisible by M) particles $\{x^{(i,j)}\}_{i=1, j=1}^{B/M, M}$ where each group share a (s^i, r^i, t^i) sample. The Monte Carlo estimate becomes

$$\hat{\mathcal{L}}_{\text{IMM}}(\theta) = \frac{1}{B/M} \sum_{i=1}^{B/M} w(s^i, t^i) \frac{1}{M^2} \sum_{j=1}^M \sum_{k=1}^M \left[k(\mathbf{f}_{s^i, t^i}^\theta(x_{t^i}^{(i,j)}), \mathbf{f}_{s^i, t^i}^\theta(x_{t^i}^{(i,k)})) + k(\mathbf{f}_{s^i, r^i}^{\theta^-}(x_{r^i}^{(i,j)}), \mathbf{f}_{s^i, r^i}^{\theta^-}(x_{r^i}^{(i,k)})) \right. \\ \left. - 2k(\mathbf{f}_{s^i, t^i}^\theta(x_{t^i}^{(i,j)}), \mathbf{f}_{s^i, r^i}^{\theta^-}(x_{r^i}^{(i,k)})) \right] \quad (67)$$

Computational efficiency. First we note that regardless of M , we require only 2 model forward passes - one with and one without stop gradient, since the model takes in all B instances together within the batch and produce outputs for the entire batch. For the calculation of our loss, although the need for M particles may imply inefficient computation, the cost of this matrix computation is negligible in practice compared to the complexity of model forward pass. Suppose a forward pass for a single instance is $\mathcal{O}(K)$, then the total computation for computation loss for a batch of B instances is $\mathcal{O}(BK) + \mathcal{O}(BM)$. Deep neural networks often has $K \gg M$, so $\mathcal{O}(BK)$ dominates the computation.

C.5. Simplified Parameterization

We derive $\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$ for each parameterization, which now generally follows the form

$$\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) = c_{\text{skip}}(s, t) \mathbf{x}_t + c_{\text{out}}(s, t) \mathbf{G}_\theta(c_{\text{in}}(s, t) \mathbf{x}_t, c_{\text{noise}}(s), c_{\text{noise}}(t))$$